Adversarial Text Purification

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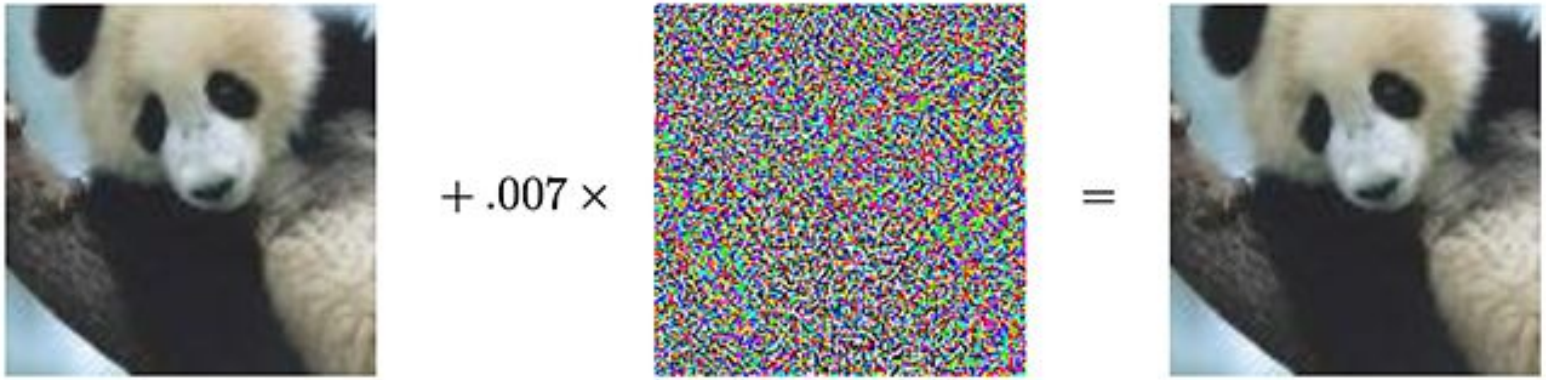
Adversarial Inputs



$\boldsymbol{x}$

"panda"
57.7% confidence

$+ .007 \times$

$\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$

"nematode"
8.2% confidence

$=$

$\boldsymbol{x} + \epsilon\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$

"gibbon"
99.3 % confidence

Defense against Adversarial Inputs

- **Adversarial Training:** We can include adversarial examples into the training process of the neural network to make the network more robust.
- **Adversarial Detection:** We can train a binary classifier to detect adversarial examples and use this on inputs before applying the neural network.
- **Adversarial Purification:** We attempt to transform the adversarial input to clean data before feeding it into our neural network, so that the prediction of the network is not affected.

Denoising Diffusion Probabilistic Models

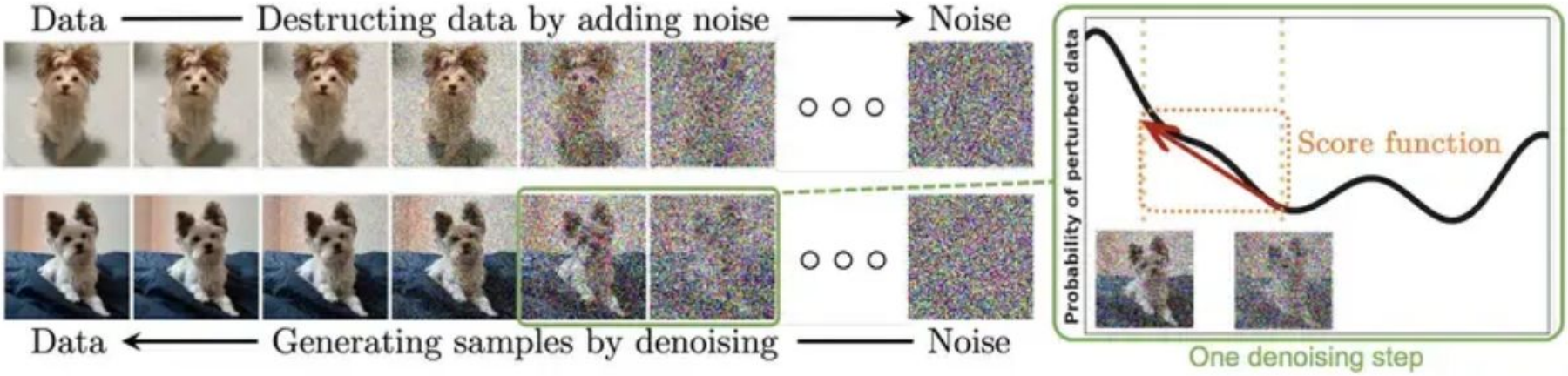
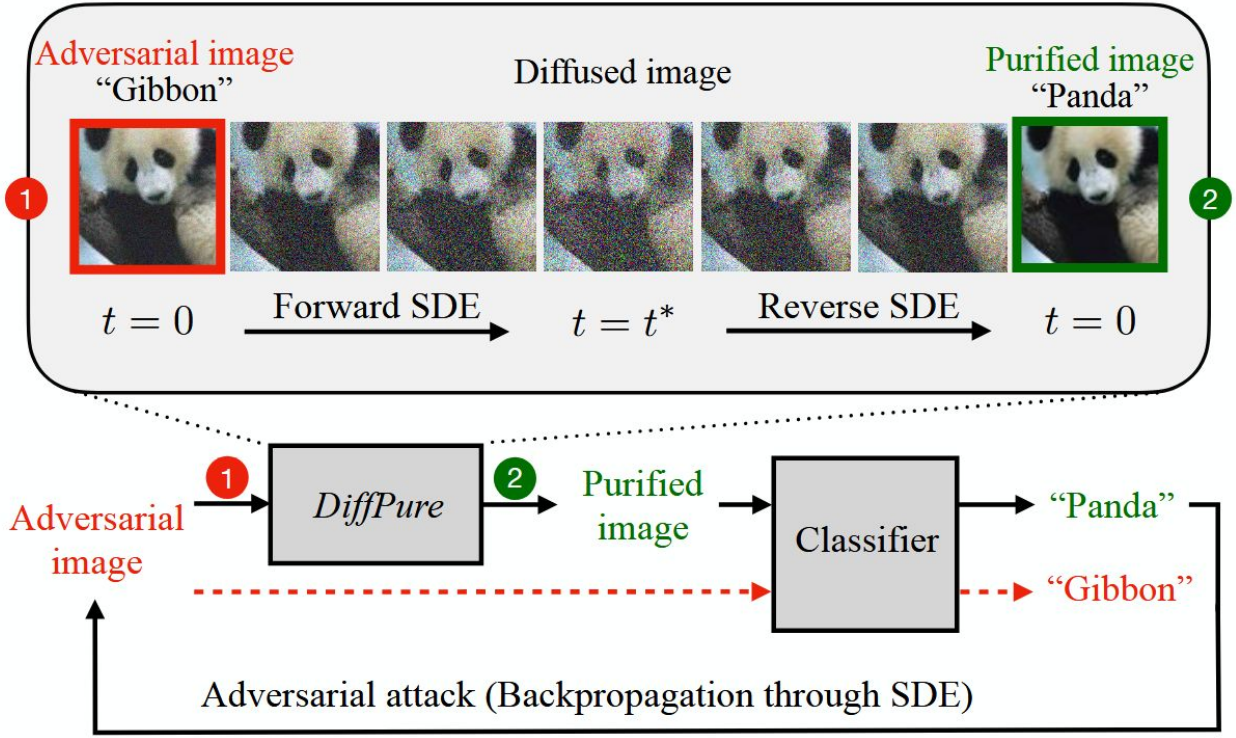


Fig. 2. Diffusion models smoothly perturb data by adding noise, then reverse this process to generate new data from noise. Each denoising step in the reverse process typically requires estimating the score function (see the illustrative figure on the right), which is a gradient pointing to the directions of data with higher likelihood and less noise.

Adversarial Purification



Adversarial Texts

	Movie Review (Positive (POS) ↔ Negative (NEG))
Original (Label: NEG)	The characters, cast in impossibly ***contrived situations***, are ***totally*** estranged from reality.
Attack (Label: POS)	The characters, cast in impossibly ***engineered circumstances***, are ***fully*** estranged from reality.
Original (Label: POS)	It cuts to the ***knot*** of what it actually means to face your ***scares***, and to ride the ***overwhelming*** **metaphorical wave** that life wherever it takes you.
Attack (Label: NEG)	It cuts to the ***core*** of what it actually means to face your ***fears***, and to ride the ***big*** **metaphorical wave** that life wherever it takes you.
	SNLI (Entailment (ENT), Neutral (NEU), Contradiction (CON))
Premise	Two small boys in blue soccer uniforms use a wooden set of steps to wash their hands.
Original (Label: CON)	The boys are in band ***uniforms***.
Adversary (Label: ENT)	The boys are in band ***garment***.
Premise	A child with wet hair is holding a butterfly decorated beach ball.
Original (Label: NEU)	The ***child*** is at the ***beach***.
Adversary (Label: ENT)	The ***youngster*** is at the ***shore***.

Adversarial Text Generation - TextFooler

- We focus on purifying adversarial text generated by word substitution attacks.
- TextFooler is a simple but strong baseline to generate adversarial text.

$$I_{w_i} = \begin{cases} F_Y(X) - F_Y(X_{\setminus w_i}), & \text{if } F(X) = F(X_{\setminus w_i}) = Y \\ (F_Y(X) - F_Y(X_{\setminus w_i})) + (F_{\bar{Y}}(X_{\setminus w_i}) - F_{\bar{Y}}(X)), & \\ \quad \text{if } F(X) = Y, F(X_{\setminus w_i}) = \bar{Y}, \text{ and } Y \neq \bar{Y}. & \end{cases} \tag{2}$$

	BERT				
	MR	**IMDB**	**Yelp**	**AG**	**Fake**
Original Accuracy	86.0	90.9	97.0	94.2	97.8
After-Attack Accuracy	11.5	13.6	6.6	12.5	19.3
% Perturbed Words	16.7	6.1	13.9	22.0	11.7
Semantic Similarity	0.65	0.86	0.74	0.57	0.76
Query Number	166	1134	827	357	4403
Average Text Length	20	215	152	43	885

Algorithm 1 Adversarial Attack by TEXTFOOLER

Input: Sentence example $X = \{w_1, w_2, ..., w_n\}$, the corresponding ground truth label Y, target model F, sentence similarity function $\text{Sim}(\cdot)$, sentence similarity threshold ϵ, word embeddings Emb over the vocabulary Vocab.
Output: Adversarial example X_{adv}

```
Initialization: X_adv ← X
for each word w_i in X do
    Compute the importance score I_{w_i} via Eq. (2)
end for

Create a set W of all words w_i ∈ X sorted by the descending
order of their importance score I_{w_i}.
Filter out the stop words in W.
for each word w_j in W do
    Initiate the set of candidates CANDIDATES by extracting
    the top N synonyms using CosSim(Emb_{w_j}, Emb_word) for
    each word in Vocab.
    CANDIDATES ← POSFilter(CANDIDATES)
    FINCANDIDATES ← { }
    for c_k in CANDIDATES do
        X' ← Replace w_j with c_k in X_adv
        if Sim(X', X_adv) > ε then
            Add c_k to the set FINCANDIDATES
            Y_k ← F(X')
            P_k ← F_{Y_k}(X')
        end if
    end for
    if there exists c_k whose prediction result Y_k ≠ Y then
        In FINCANDIDATES, only keep the candidates c_k whose
        prediction result Y_k ≠ Y
        c* ← argmax_{c∈FINCANDIDATES} Sim(X, X'_{w_j→c})
        X_adv ← Replace w_j with c* in X_adv
        return X_adv
    else if P_{Y_k}(X_adv) > min_{c_k∈FINCANDIDATES} P_k then
        c* ← argmin_{c_k∈FINCANDIDATES} P_k
        X_adv ← Replace w_j with c* in X_adv
    end if
end for
return None
```

Adversarial Text Purification

- Despite the success of adversarial purification in the field of computer vision, owing to generative models such as diffusion models, its is relatively unexplored in the field of natural language processing due to the discrete nature of the text inputs.
- Any perturbation to a token completely changes its nature and can change the entire text meaning.

She gave her cat food → She gave her cat away

Diffusion Language Models

- Previous research proposed that by using masked language models like BERT, we can inject noise by masking input texts and reconstruct the masked texts.
- In this way, we construct an adversarial purification process for textual models against the most widely used word-substitution adversarial attacks.
- But with recent advancements in generative capabilities, we can use more sophisticated architectures like masked diffusion models.
- Autoregressive models do left-to-right generation, whereas Diffusion language models allows for infilling.

Score Entropy Discrete Diffusion model

- Similar to denoising diffusion models, we use a diffusion scheduler to add noise by masking tokens.
- And using concrete score ratios, we sample a less noised version of the input.
- We can use the SEDD model to purify text by iterative denoising of the masked adversarial text.

study ants bear burrito skyline song

Results

OG : Surrey poised to sign harbhajan surrey are waiting for approval from the board of control for cricket in india before announcing harbhajan singh as an overseas signing for 2005 (Sports)

Adv : Surrey mortgages to sign harbhajan surrey are waiting for approval from the board of controlled for grasshopper in india before announcing harbhajan singing as an overseas firma for 2005 (Business)

Masked: Surrey mortgages to sign harbhajan [MASK] are [MASK] for approval from the [MASK] of controlled for [MASK] [MASK] india before [MASK] harbhajan singing as an [MASK] [MASK] for 2005

Purified: Surrey mortgages to sign harbhajan Oprey are waiting for approval from the coaches who controlled for flameshopper's india before seeing harbhajan singing as an Olympic hurricanea in 2005 (Sports)

Results (Classification Accuracy)

	Original Text	Adversarial Text	Purified Original Text	Purified Adversarial Text
AG News	94.2	12.5	89.61	68.3
IMDB	90.9	13.6	83.62	55.09

Ablation Study on successful adversarial examples

Timestep	0.1	0.2	0.3	0.1&0.2	0.2&0.3	0.1&0.3	All
AG News	0.539	0.657	0.653	0.382	0.471	0.381	0.652
IMDB	0.507	0.522	0.498	0.467	0.448	0.446	0.506

Ablation Study on successful adversarial examples at t=0.2

Outer loop iterations	Accuracy
1	0.657
2	0.6475
5	0.6489
10	0.6539

Further Work

- BERT classification task has become obsolete nowadays. Further work can focus on purifying texts like adversarial prompts or jailbreaking attacks.
- SEDD is just the introduction of diffusion models in text. It only has comparable performance of GPT-2. Currently there are many new diffusion models emerging like LLaDa (Large Language Diffusion Models) or Dream-7B.
- Current diffusion models add noise via masking. But there can be better approaches which can add noise to tokens so that the new token is semantically similar to the original token.
- https://docs.google.com/presentation/d/1EHRMpcG3NGy2S67x3rcbSU6NUppddnF7B-DCIyr12oM/edit?usp=sharing